

Second Partial

Fco. Vazquez

Tec de Monterrey

Rule of derivatives

Power rule

$$\frac{d}{dx}x^n = nx^{n-1}$$

Chain rule

$$\frac{d}{dx}u(v(x)) = u'(v(x)) \cdot v'(x)$$

Quotient rule

$$\frac{d}{dx} \frac{u(x)}{v(x)} = \frac{u'v - uv'}{v^2}$$

Product rule

$$\frac{d}{dx}u(x)v(x) = u'v + uv'$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

Quotient rule

The function $f(x)$ is a quotient of functions, where $u = x$ and $v = (x^2 + 1)^{3/2}$. By applying the quotient rule, $\frac{d}{dx}u/v = \frac{1}{v^2}(u'v - uv')$, we get,

$$\begin{aligned} f'(x) &= \frac{1}{\left[(x^2 + 1)^{3/2}\right]^2} \left[\left(\frac{d}{dx}x \right) (x^2 + 1)^{3/2} - x \left(\frac{d}{dx}(x^2 + 1)^{3/2} \right) \right] \\ &= \frac{1}{(x^2 + 1)^3} \left[\left(\frac{d}{dx}x \right) (x^2 + 1)^{3/2} - x \left(\frac{d}{dx}(x^2 + 1)^{3/2} \right) \right] \end{aligned}$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

$$f'(x) = \frac{1}{(x^2+1)^3} \left[\left(\frac{d}{dx} x \right) (x^2+1)^{3/2} - x \left(\frac{d}{dx} (x^2+1)^{3/2} \right) \right]$$

Power rule: $\frac{d}{dx} x$

$$\frac{d}{dx} x = 1x^{1-1} = 1x^0 = 1 \cdot 1 = 1$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

Replace

$$\begin{aligned} f'(x) &= \frac{1}{(x^2+1)^3} \left[\left(\cancel{\frac{d}{dx} x} \right)^1 (x^2+1)^{3/2} - x \left(\frac{d}{dx} (x^2+1)^{3/2} \right) \right] \\ &= \frac{1}{(x^2+1)^3} \left[1 \cdot (x^2+1)^{3/2} - x \left(\frac{d}{dx} (x^2+1)^{3/2} \right) \right] \end{aligned}$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

Chain rule: $\frac{d}{dx}(x^2 + 1)^{3/2}$

Here, $u = x^{3/2}$ and $v = x^2 + 1$, so that $(u \circ v)(x) = (x^2 + 1)^{3/2}$. With this, we can apply the chain rule, $\frac{d}{dx}u(v(x)) = u'(v(x)) \cdot v'(x)$,

$$\begin{aligned}\frac{d}{dx}(x^2 + 1)^{3/2} &= \left. \frac{d}{dx}x^{3/2} \right|_{x=x^2+1} \cdot \left[\frac{d}{dx}(x^2 + 1) \right] \\ &= \left. \frac{3}{2}x^{1/2} \right|_{x=x^2+1} \cdot [2x + 0]\end{aligned}$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

Chain rule: $\frac{d}{dx}(x^2+1)^{3/2}$

$$\begin{aligned}\frac{d}{dx}(x^2+1)^{3/2} &= \frac{3}{2}x^{1/2} \Big|_{x=x^2+1} \cdot [2x+0] \\ &= \frac{3}{2}\sqrt{x^2+1} \cdot 2x \\ &= 3x\sqrt{x^2+1}\end{aligned}$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

Replace

$$\begin{aligned}
 f'(x) &= \frac{1}{(x^2+1)^3} \left[1 \cdot (x^2+1)^{3/2} - x \left(\frac{d}{dx} (x^2+1)^{3/2} \right) \right] 3x\sqrt{x^2+1} \\
 &= \frac{1}{(x^2+1)^3} \left[1 \cdot (x^2+1)^{3/2} - x \left(3x\sqrt{x^2+1} \right) \right]
 \end{aligned}$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

Simplify

$$\begin{aligned} f'(x) &= \frac{1}{(x^2+1)^3} \left[1 \cdot (x^2+1)^{3/2} - x \left(3x\sqrt{x^2+1} \right) \right] \\ &= \frac{1}{(x^2+1)^3} \left[(x^2+1)^{3/2} - 3x^2\sqrt{x^2+1} \right] \end{aligned}$$

$$f'(x) = \frac{1-2x^2}{\sqrt{(x^2+1)^5}}$$

Class Activity

Compute the derivative of the following functions:

$$f(x) = \frac{2}{1 + e^{-2(x-4)}}$$

$$g(x) = \frac{2x}{(x^2 + 4)^{3/2}}$$

$$f'(x) = \frac{4e^{-2(x-4)}}{[1 + e^{-2(x-4)}]^2}$$

$$g'(x) = \frac{8 - 4x^2}{\sqrt{(x^2 + 4)^5}}$$

Power rule: $\frac{d}{dx} x^n = nx^{n-1}$

Chain rule: $\frac{d}{dx} u(v(x)) = u'(v(x)) \cdot v'(x)$

Quotient rule: $\frac{d}{dx} \frac{u(x)}{v(x)} = \frac{u'v - uv'}{v^2}$

Product rule: $\frac{d}{dx} u(x)v(x) = u'v + uv'$